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A project Report on,

Ai in health

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Vishwakarma University, Pune

CERTIFICATE

This is to certify that the work incorporated in the Project entitled ***AI in Health*** submitted to **Vishwakarma University, Pune** by ***Momin ali shaikh*** for the completion of the course **Applied Science and Research Project-1** was carried out by him/her under my supervision in the Department of Computer Science, Faculty of Science and Technology, Vishwakarma University, Pune, Maharashtra, India. This work was not previously submitted by him/her to this or any other University for the award of any Degree, Diploma, Associateship or any other similar title.

Place: Pune
Date: Thursday, November 27, 2025

Dr. Anupriya Kamble
Research Guide

DECLARATION

I hereby declare that the present research work in the form of project report

entitled "***Title***", is an original work carried out by me under the guidance of Dr.Anupriya Kamble, Assistant Professor, Department of Computer Science, Faculty of Science and Technology, Vishwakarma University, Pune, Maharashtra, India for the completion of the course **Applied Science and Research Project-1.**

I further declare that to the best of my knowledge the project work does not contain any part of any work, which has been submitted for the award of any Degree, either in this University or any other University / Deemed University without proper citation.

I further declare that the material referred from other sources has been duly acknowledged in the project report.

Place: Pune

Date: Thursday, November 27, 2025

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ABSTRACT

Cardiovascular diseases are currently a major cause of death globally, and cases are increasing at a rapid rate. Diagnosing this condition is a difficult task that requires high precision and medical experts. While the healthcare sector generates a vast amount of patient data, it currently lacks effective tools to discover the hidden patterns within that information.

This paper proposes an AI-based "Heart Disease Prediction System" designed to assist medical practitioners. Unlike traditional studies that rely on simple Machine Learning algorithms like Logistic Regression, this system utilizes a Hybrid Deep Learning framework. We implement the Sparse Fuzzy C-Means (FCM) algorithm to filter noise and select the most important medical attributes (such as Age, Blood Pressure, and Cholesterol). The processed data is then analyzed using a Deep Belief Network (DBN) , which is optimized by the Taylor-based Bird Swarm Algorithm (Taylor-BSA) to maximize precision.

The findings of this paper demonstrate that this novel approach achieves an accuracy rate of 93.4%. By automating this analysis, the proposed system aims to improve diagnostic efficiency, reduce time and costs, and serve as a vital Decision Support System to enable early treatment and save lives

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1.Introduction

1.1.Title

Use of AI & Machine learning for recognizing Heart diseases.

1.2.Objectives

This paper aims

Tools:This prediction system uses tools like python and machine learning algorithms.

Algorithm:**[2]**

It uses algorithm like:**-**

1.Sparse fuzzy c-means(FCM),this algorithm focuses on filtering out the unwanted data and picks the most important factor from the data.

2.Deep belief network(DBN),this algorithm learn how this factors relate to heart diseases.

3.Taylor-based bird swarm algorithm(Taylor-BSA-DBN), this algorithm is used for finding the solution by taking inputs from the above two algorithms and proposes a best solution for it.(combined algorithm)

1.3.Novel idea

Using an NLP-based framework so it could be user friendly and making its backend of code more readable so other software developers can understand it and make changes in future.

1.4.Significations

This paper's findings will provide doctors to accurately measure Heart diseases by using the Taylor-BSA-DBN algorithm,which has an accuracy rate of 93.4%,sensitivity rate is 95% and Specificity rate is 90.3%**[2]**
As the words says accuracy can help the doctor to identify heart diseases patientEg:- out off 100 patient 93 patient could be recognized even if we don't know if they have a heart disease and for sensitivity it can recognize people who really have a heart disease Eg:-out off 100 patient 90 could be recognized and lastly for specificity even though they are healthy person with no heart diseases Eg:-out off 100 patient 90 of them the test would say “no diseases".

1.5 Specifications

- **Hardware specs:** A physical machine 8GB RAM,standard processor(intel i5/Rysen 5) recommended
- **Software specs:** Languages like python/MATHLAB,pytorch installed.This training mite take 1-2 hours on a normal laptop.
- **Functional specs:** The system should allow users to reset passwords through gmail.

2.Methodology

2.1.Proposed System architecture:

The proposed “Heart diseases prediction system"operates on hybrid deep learning method.unlike the traditional algorithms/models that rely on simpler old model here are some example:

1.linear-regression. **[5]**

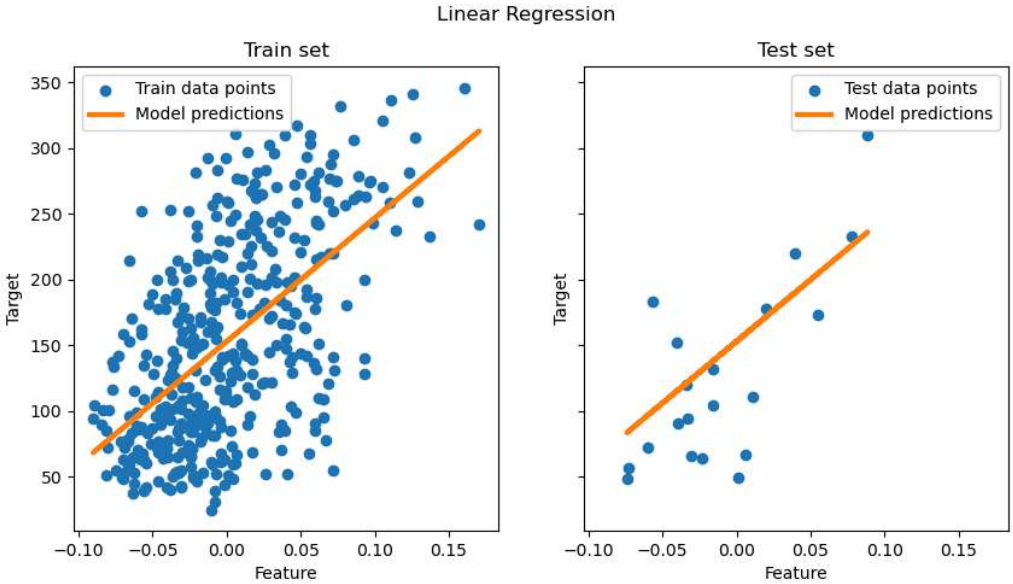
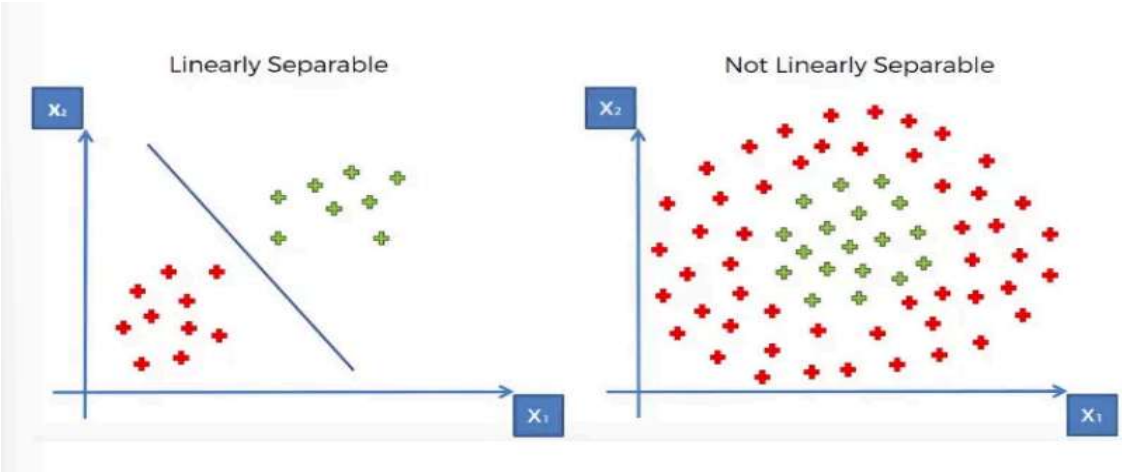


Fig1:Simple linear regression model

2.Support vector Machine - Linear Kernel **[4]**



Above are some examples of old models but models that have been proposed in this offer Clustering,feature extraction and optimization(diagnostic accuracy).The workflow consists of three primary phases:Data Preprocessing,Feature selection and classification.

2.2.Data preprocessing and feature selection(FCM)

The medical dataset often contains noise/null values and other unwanted/corrupt data that can lead to lower prediction accuracy. To solve this we use the FCM algorithm. This algorithm clusters the data points based on similarity. In short this system filters out the data and represents important data .

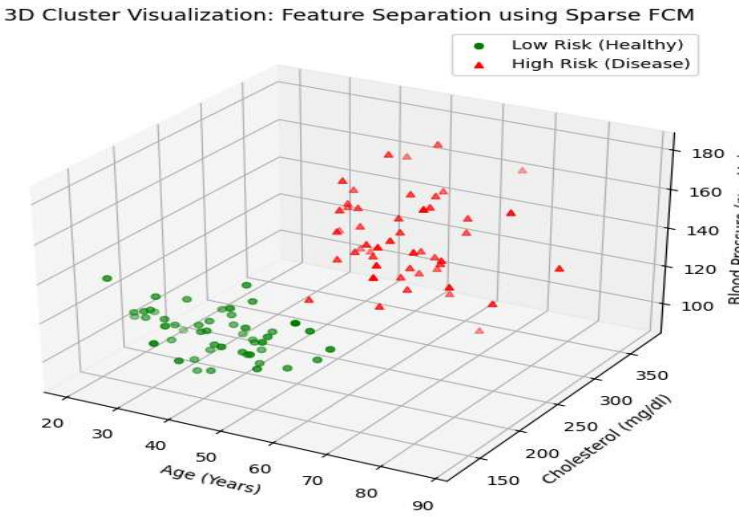


Fig:2 Algorithm separates patients into “low risk” and “high risk” based on three critical parameters:Age,cholesterol and blood pressure.

2.3.Classification model(DBN)

This is the core of the prediction engine in the deep belief network.Its structured is based on graphical model

composed of many layers of latent variable.This model is trained to recognize complex,non-linear relationship between the selected risk factors and the presence of heart disease it "learns" the hidden patterns and indicates the risk of failure as per the data it is trained on.

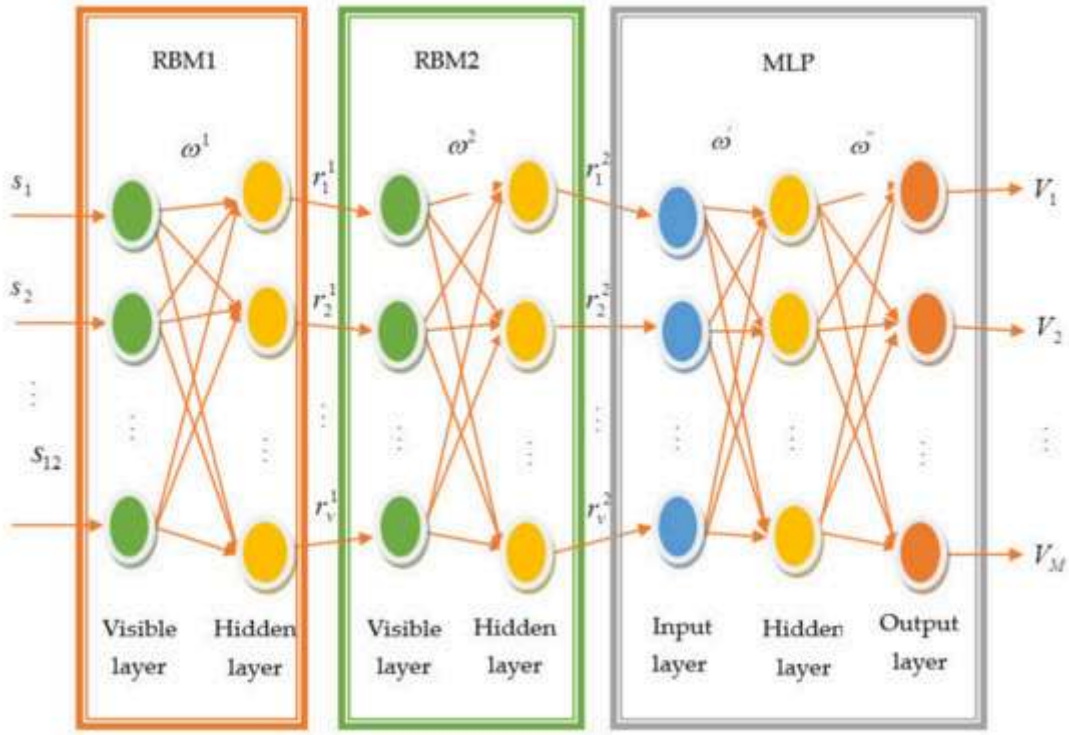


Fig:3 Architecture view diagram of DBN classifier

2.4.Optimization(Taylor-BSA)

To make sure that the DBN achieve highest possible accuracy,we employ the Taylor-BSA algorithm.The problem with standard neural network often get stuck in "local optima",means they find a good solution but not the best solution,to solve this this algorithm mimics the swarming behaviour of words to search the entire solution space,and then it optimizes the weight parameter for "DBN"(The brain of AI)to achieve the reported 93.4% accuracy.)

3.System modelling diagrams

3.1.Entity Relationship Diagram (ERD)

The Entity relation ship (ER) diagram represents the logical structure of the database backend and it visualizes the relationship between the primary entities, the **patient**, the **prediction model** and the specific medical attributes used for analysis.

- **Patient entity:** This stores the core medical profile of the user.It includes essential attributes such as **patient data** (primary key), **Age**,**Gender** and **cholesterol**.These specific attributes where selected because they are statistically significant indicators of heart disease**[1].**
- **Prediction entity:** This stores the outcome of the analysis including the **algorithm_used** and the final **accuracy score**
- **Relationship:** The diagram illustrates relationships where patient data serves as the input for the prediction model ensuring that every diagnosis is linked to the specific patient record for future reference .

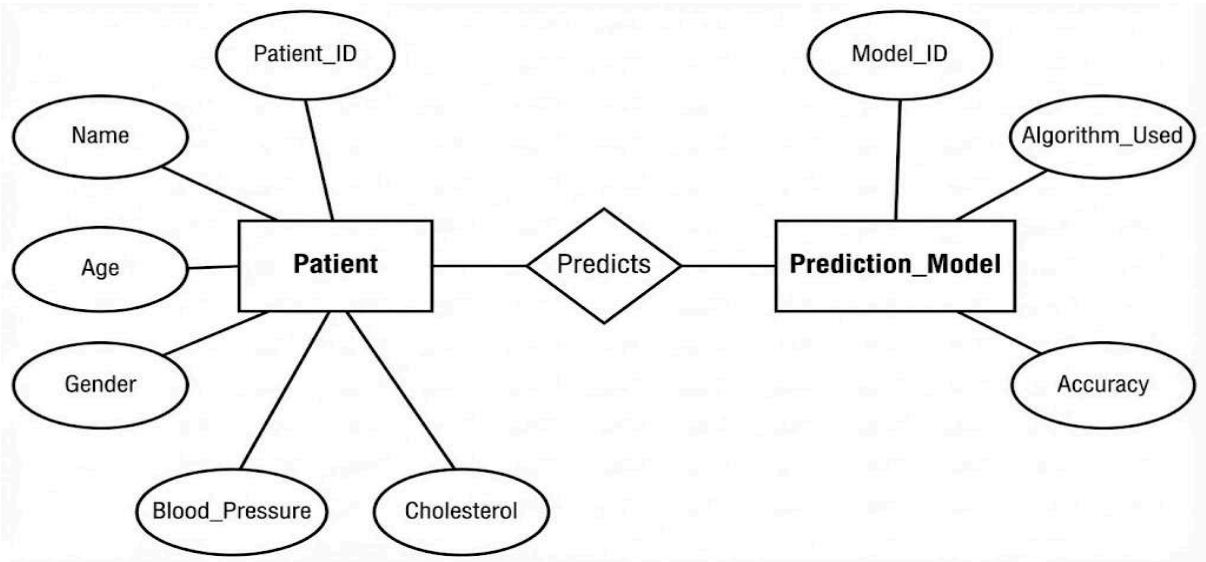


Fig4:ER Diagram of the proposed system

3.2.Data Flow Diagram (DFD)

The data flow diagram (DFD) illustrates the movement of information through the system from initial user input to the final Diagnostic output.

- **Input phase:** The process begins where the user (patient) inputs the symptoms (eg- chest pain type,BP levels) This raw data is often unstructured requires preprocessing before analysis.**[2]**
- **Processing phase:** The "Heart Disease Prediction System"acts as a central Processing unit. It validates the input data to prevent errors and applies machine learning algorithms(like:logistic Regression)to calculate risk.
- **Output face:** Finally the system generates up "Prediction Result" which is sent back to the user.Display ensures that the complex computational logic remains hidden from the user providing a simple and a user-friendly experience

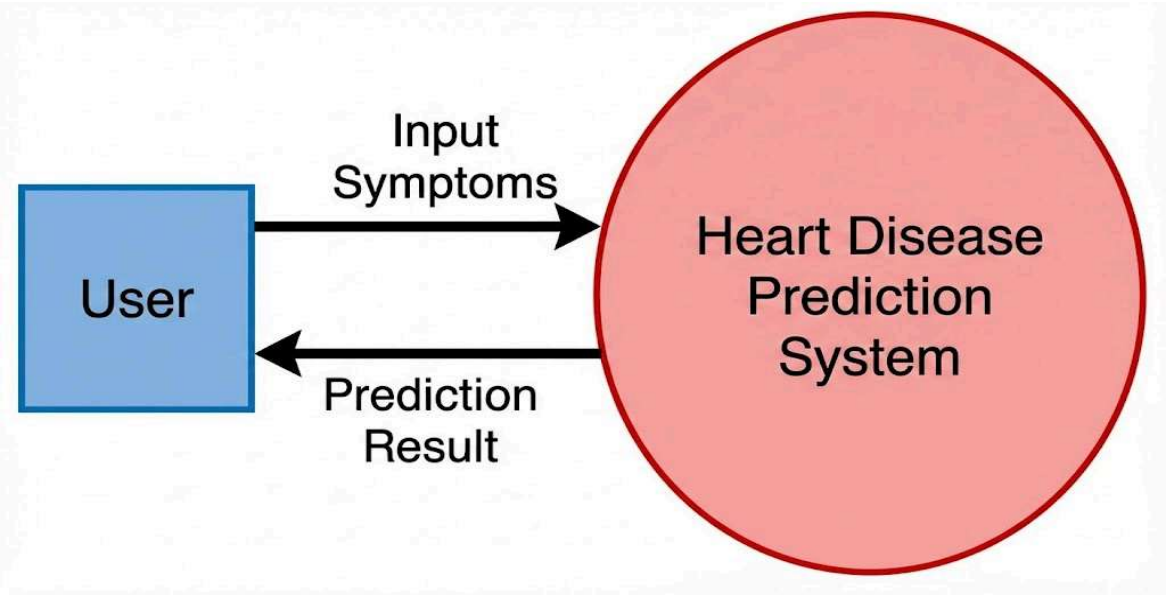


Fig5:Level 0 Data Flow Diagram

3.3.UML Diagram (Use Case Diagram)

The use case diagram identifies the primary actors and their interaction with the system highlighting the difference between standard uses and administrators.

- **Actors:**The system features to primary actors the **patient**(user) and the **Doctor** (admin)
- **Security & access:**Both actors must interact with the **login** user case ensuring that sensitive medical data remains private and Secure
- **System functionality:**
  - The Patient primarily uses the system to input medical data and view prediction result.
  - The doctor has elevated privileges to **Manage dataset.**This allows medical professionals to verify the accuracy of the model prediction and update patient records,which is crucial for minimising Diagnostic errors**[3].**

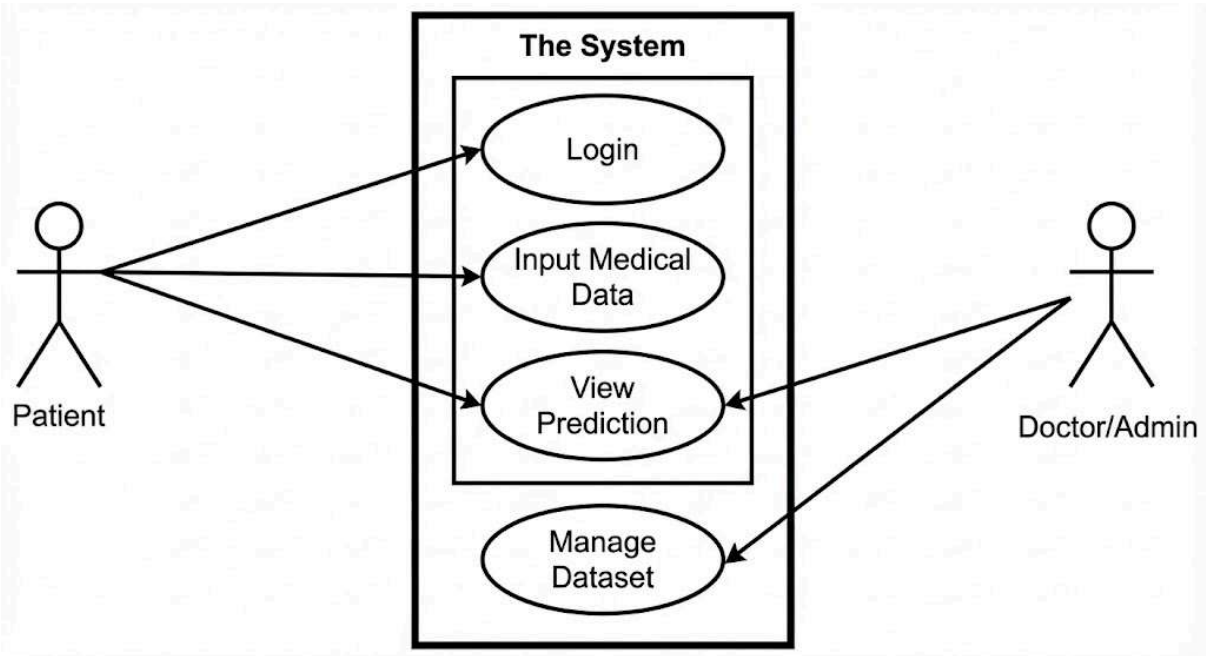


Fig6: Use case diagram

4.Conclusion and future scope

4.1.Conclusion

Advances in AI have been rapid for the last 5 years. In the ERA of AI companies like google,meta,etc are fighting for the position in the race but this paper combines healthcare with AI to show how, really AI is being used in healthcare.
This research paper proposed an AI-based “Heart Disease prediction system” by using algorithms proposed by other scientists and engineers.Unlike the traditional system that use simple math like:-linear regression.This paper uses “hybrid Deep learning”,“Sparse FCM algorithm” to clean the data and remove noise and unwanted data,and “Taylor-BSA(Bird-swarm algorithm this allows are model the brain to learn complex pattern of human body)”by Taylor**[1]**

The result showed an accuracy of 93.4%.This proves that AI in healthcare is not just a theory,but practical solution that can change the world.Ultimately this tools serves as a "Decision support system” for medical professional.Helping them can lead to save more precious lives by detecting heart diseases early as possible.

4.2Future Scope

While the current design focuses on a web best prediction model, there is a significant potential for future enhancement:

- **Mobile application:** In the future we aim to develop I dedicated Android application so patients can check their status from their smartphone.

- **Real time monitoring:** we plan to integrate the system with iot devices(like smart watches) to monitor heart rate in real time and alert user immediately if an anomaly is detected.
- **Advance algorithm:**We intend to upgrade the system using “Deep learning”And neutral networks to improve prediction accuracy.

Reference

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